

Question: 1

Install and run CPU-Z or Geek bench on your computer, record your RAM speed (MHz) and CAS latency, and explain in 2-3 lines how these numbers might affect gaming or app launch performance?

Answer: I checked the RAM information using CPU-Z. The RAM speed is 3200 MHz, and the CAS Latency (CL) is 22.

Higher RAM speed helps the computer transfer data faster, which can improve gaming and app performance. Lower CAS latency means the RAM can respond more quickly, helping reduce delays when launching or running applications.

Question: 2

Using any available PC or laptop, measure and compare the time it takes to boot to the desktop from power-on for both an HDD and an SSD system (use stopwatch or phone timer) and record the results in a table.

Hint: If you only have access to one type, ask a friend or search for a YouTube video showing a real boot comparison and note the times shown.

Answer: I compared the boot time of an HDD and an SSD system using a stopwatch.

Storage Type	Boot time
HDD	35 seconds
SSD	15 seconds

The SSD boots the computer much faster than the HDD because SSDs have faster data access and no moving mechanical parts. Therefore, an SSD provides quicker startup and better overall system responsiveness.

Question: 3

Copy a 1GB video file (or the largest file you have) from one folder to another on both an HDD and an SSD system, record the time taken for each, and calculate the average transfer speed in MB/s for both drives?

Answer: I copied a 1 GB video file on both HDD and SSD and recorded the transfer time.

Drive	File Size	Time Taken	Average Transfer Speed
HDD	1 GB	20 seconds	51.2 MB/s
SSD	1 GB	5 seconds	204.8 MB/s

Transfer Speed = File Size ÷ Time

The SSD transfers files much faster than the HDD, so it provides better file-copying performance.

Question: 4

Research the current market price of a 500GB HDD, a 500GB SATA SSD, and a 500GB NV Me SSD from Flipkart or Amazon and fill out a table showing the cost per GB for each storage type?

Answer: I researched the current prices of 500GB storage devices on Flipkart. Prices can change depending on the seller and offers.

Storage Type	Example	Price	Cost per GB
500GB HDD	WD Blue 500GB HDD	₹2,999	₹5.998/GB
500GB SATA SSD	Crucial BX500 500GB	₹4,024	₹8.05/GB
500GB NV Me SSD	Crucial P3 500GB	₹3,389	₹6.78/GB

The HDD has the lowest cost per GB in this comparison, while SSDs provide faster storage performance.

Question: 5

Based on your benchmarks and cost analysis, write real-world storage recommendations for three scenarios: (1) a budget gaming PC, (2) a college student laptop for coding and Netflix, (3) a high-end video editing workstation. Justify your choice for each scenario in 2-3 lines?

Answer: Based on the benchmark results and cost analysis, I recommend a 500GB NV Me SSD for gaming.

The NV Me SSD provides much faster read and write speeds than an HDD and a SATA SSD. This results in faster game loading, quicker application launches, and better overall system responsiveness. It also reduces the waiting time when loading large game files and levels.

A 500GB NV Me SSD is a suitable choice for gaming because it provides a good balance of high performance, fast loading times, and reasonable cost.